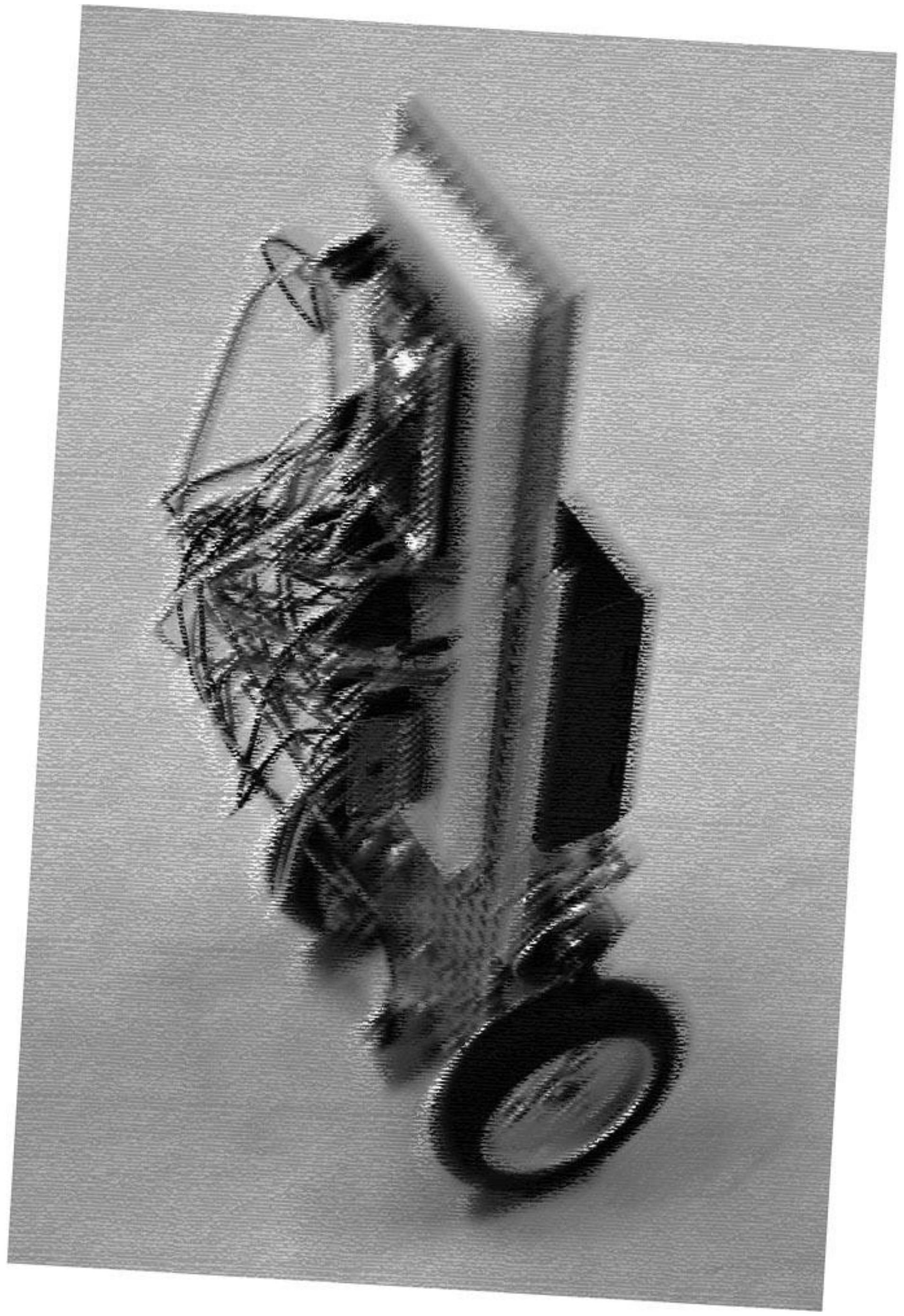
PRACTICAL EXERCISE · WEEK 2

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE	Department of Precision Engineering · 2026 S1S2
SESSIONS	Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14
TODAY	Course Introduction — 10 minutes



THE FOUR-WEEK ARC

Timeline

1

Week 1
Standing

Assembly & bring-up

Angle-only **standing** demo

2

Week 2
Position control

Build a rotary encoder

Position control

Simulation in Simulink

3

Week 3
EXTENSION

Load & disturbance tests

Advanced control (LQR/MPC)

……

4

Week 4
presentation

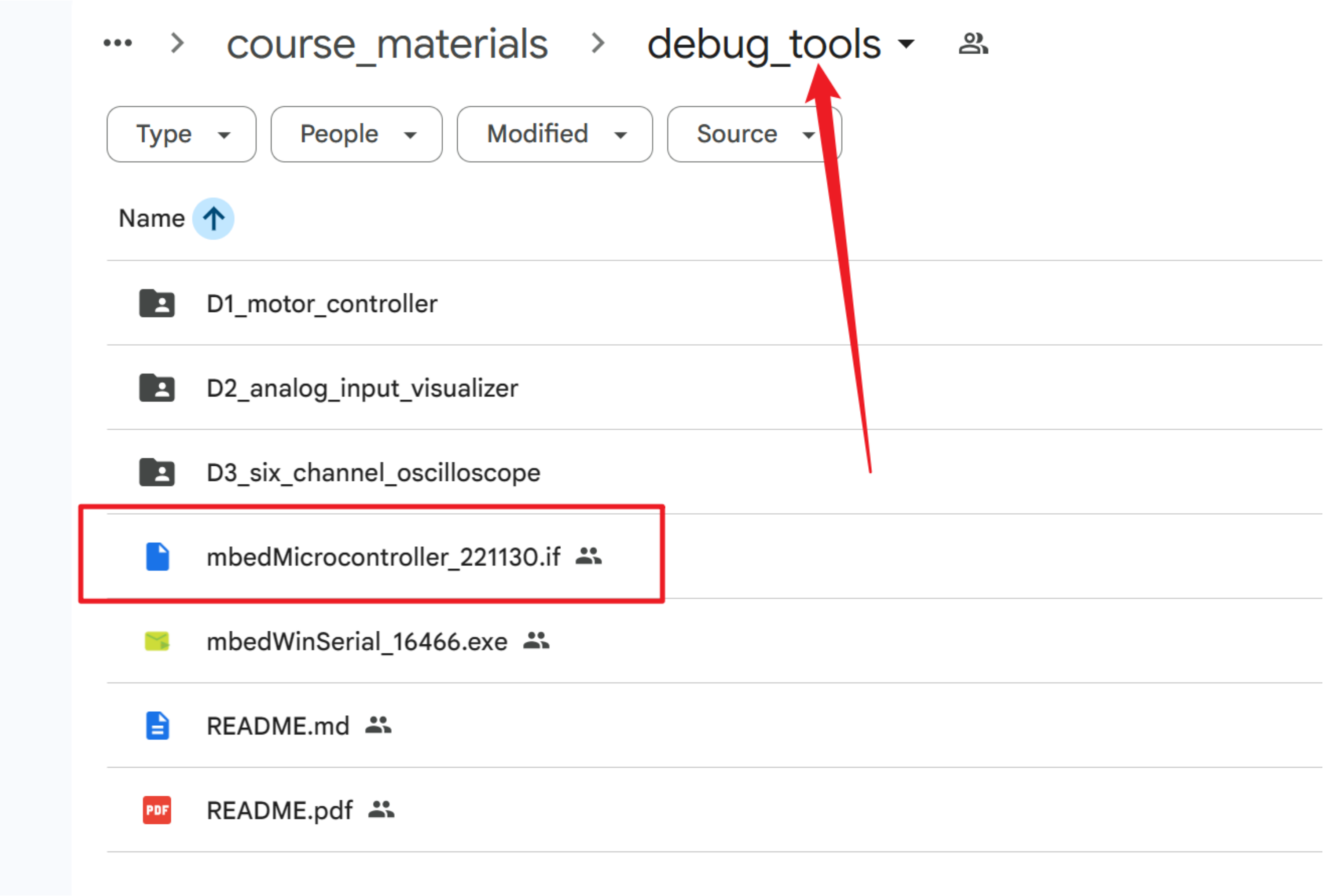
10-minute presentation

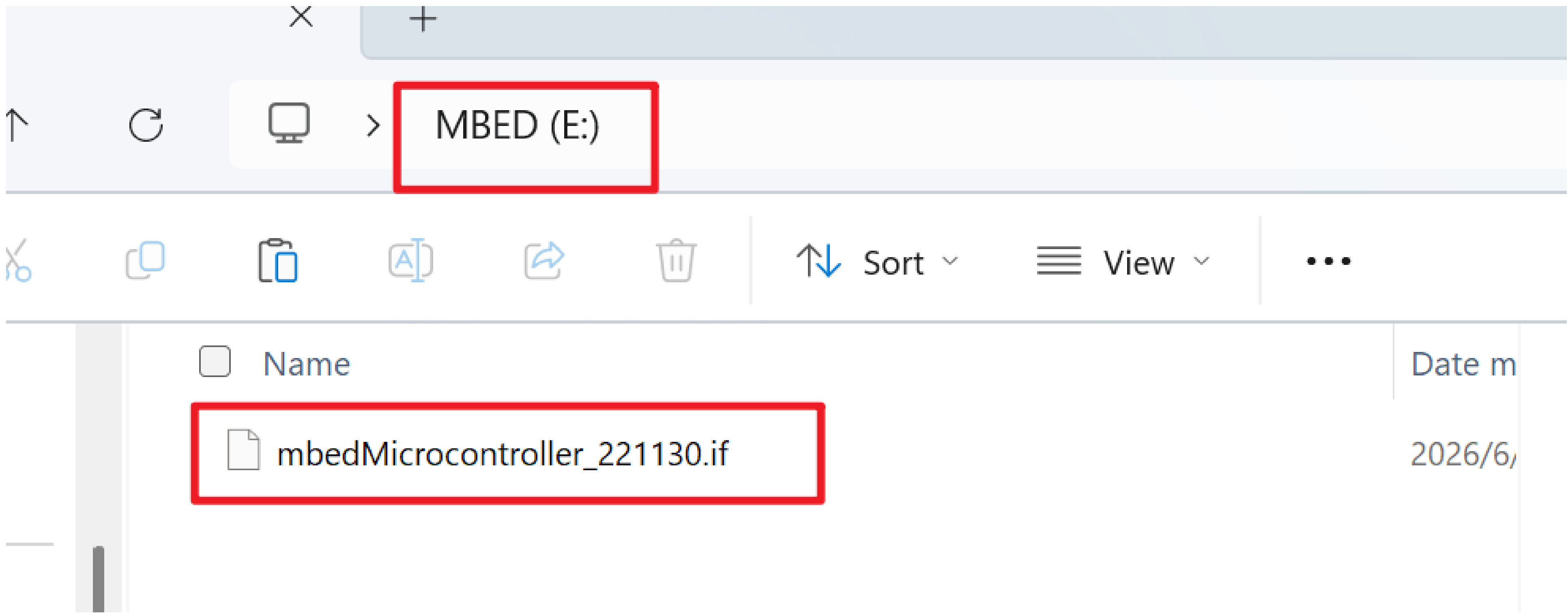
Live demo & Q&A

Final grade · 40%

Week 3 is for extended tasks, or you can continue working on the tasks of Week 1 and Week2

If you are unable to connect to the Windows serial port, you can try updating the mbed firmware.





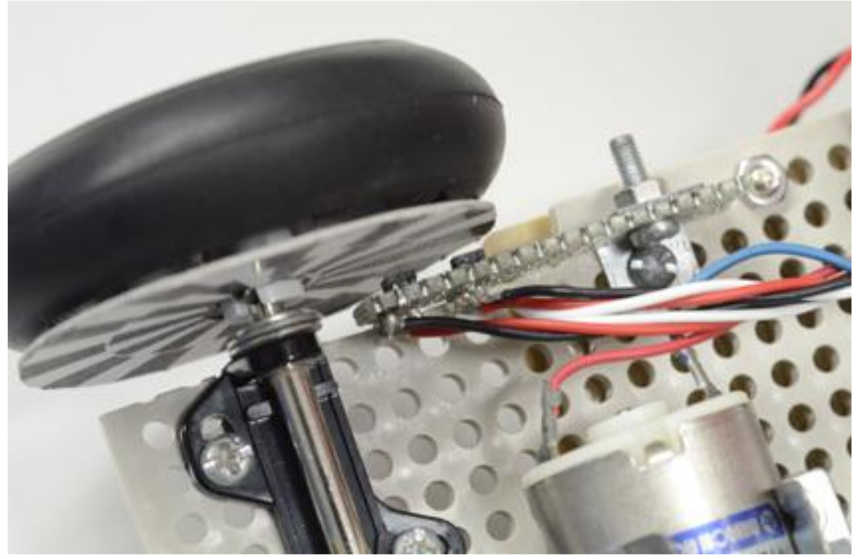
Week 2

Assign the tasks among yourselves.

A TASK A

STUDENT A · Hardware

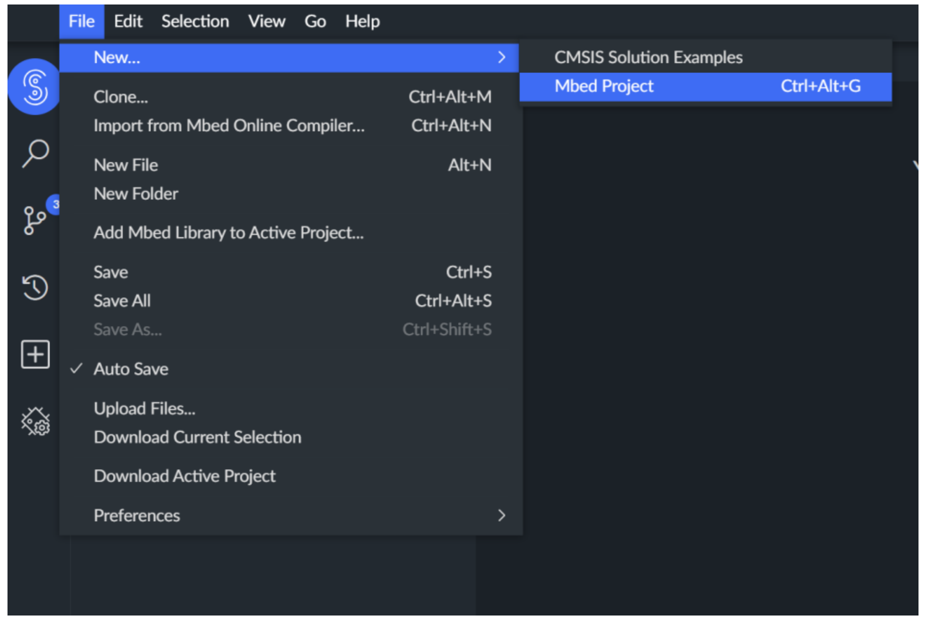
Encoder Fabrication



B TASK B

STUDENT B · FIRMWARE

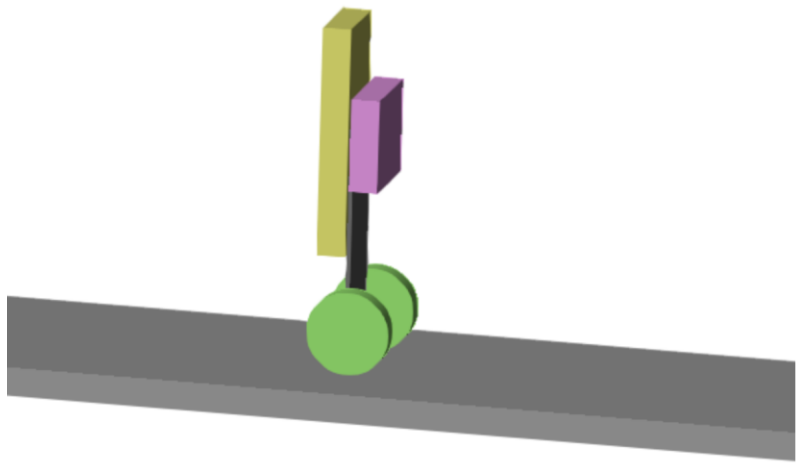
Add Encoder Feedback to Microcontroller



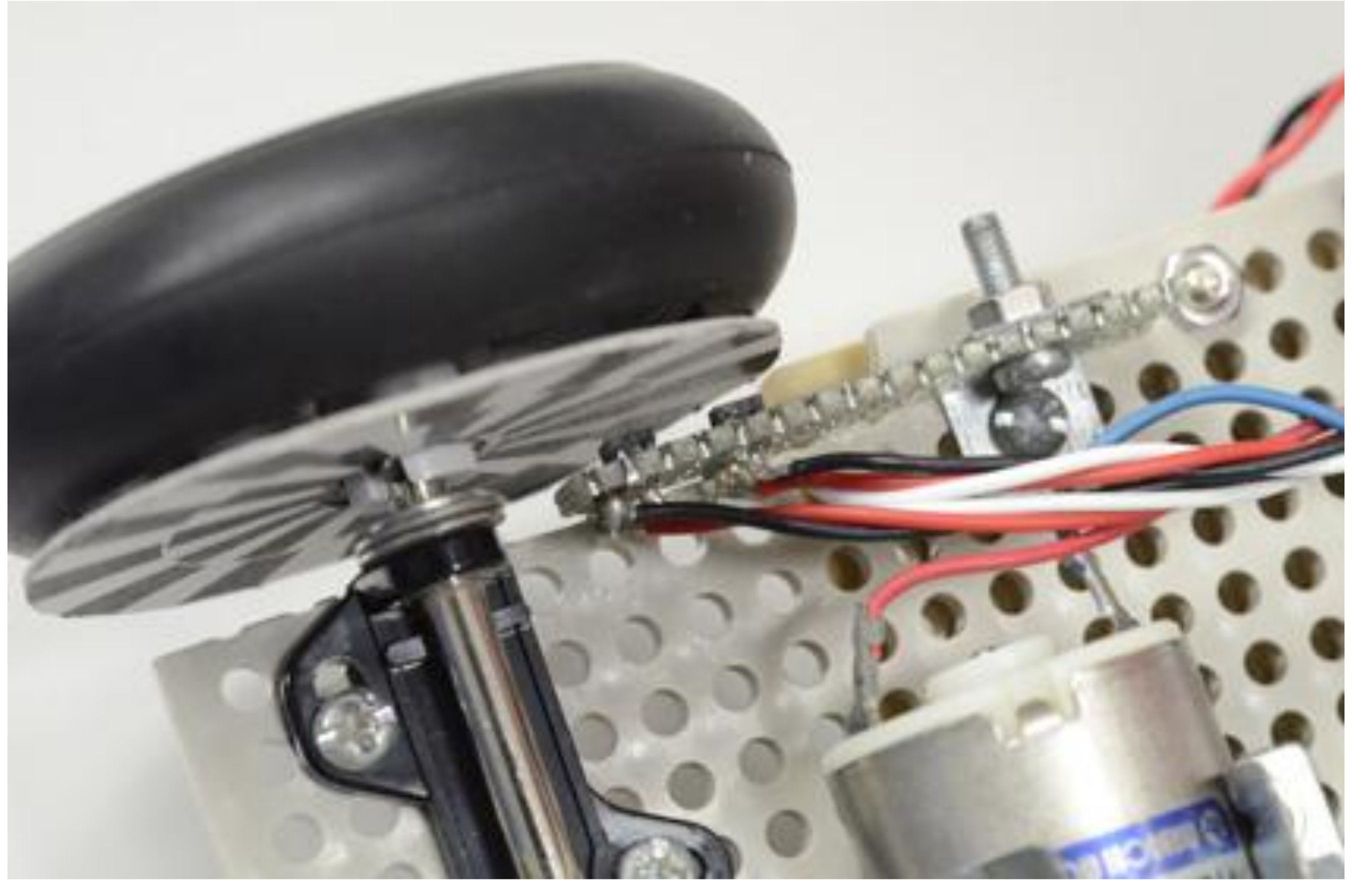
C TASK C

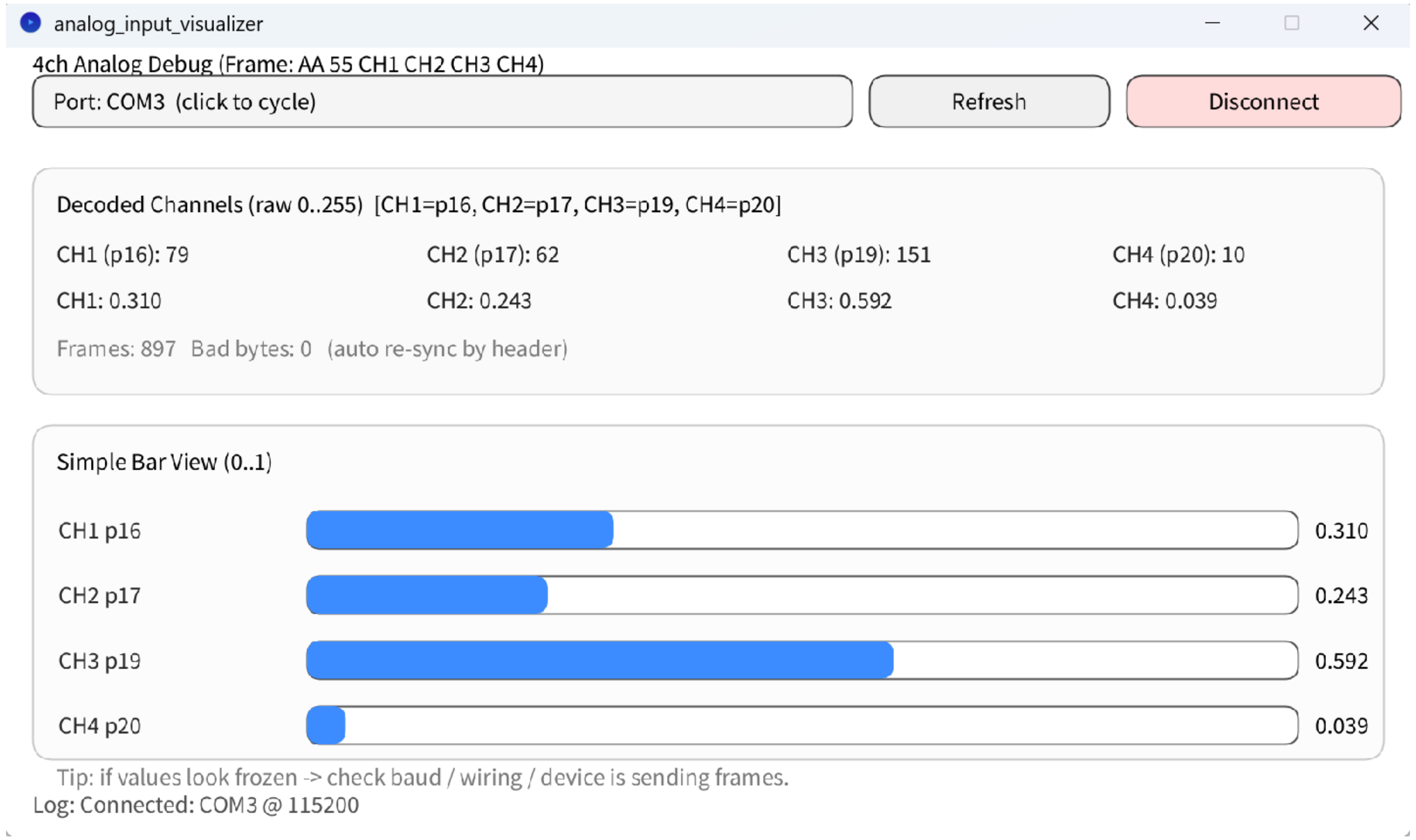
STUDENT C · SIMULATION

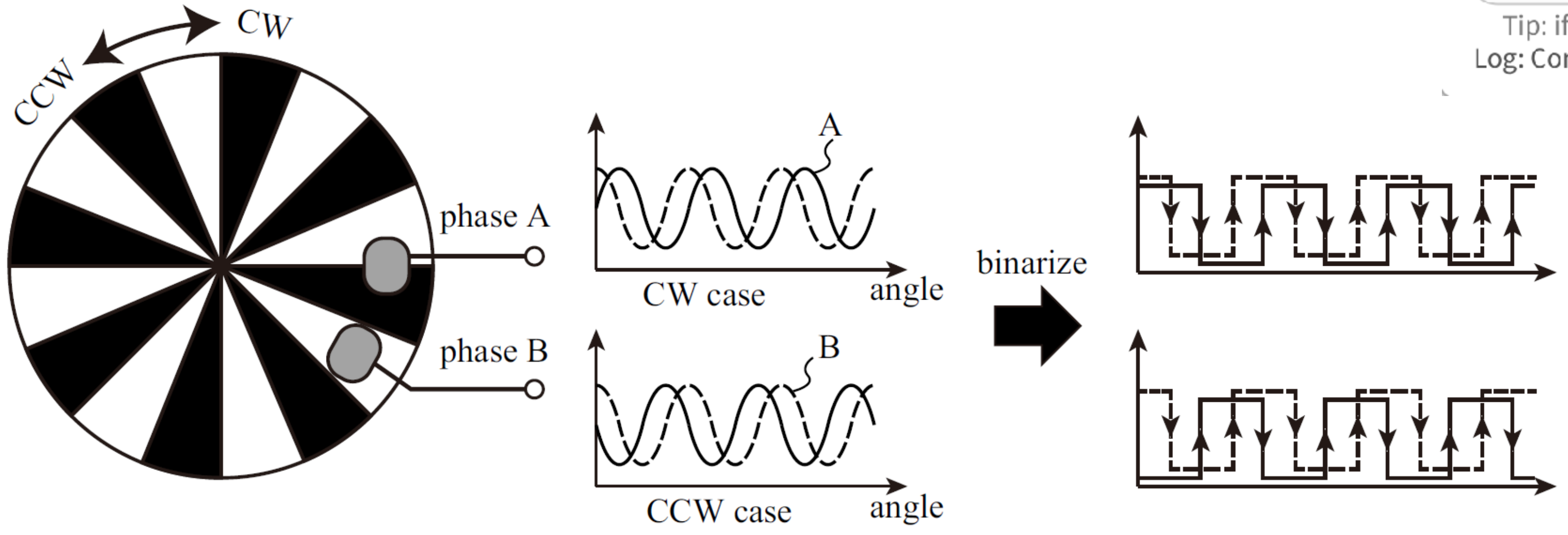
Simulink Simulation



Encoder Fabrication



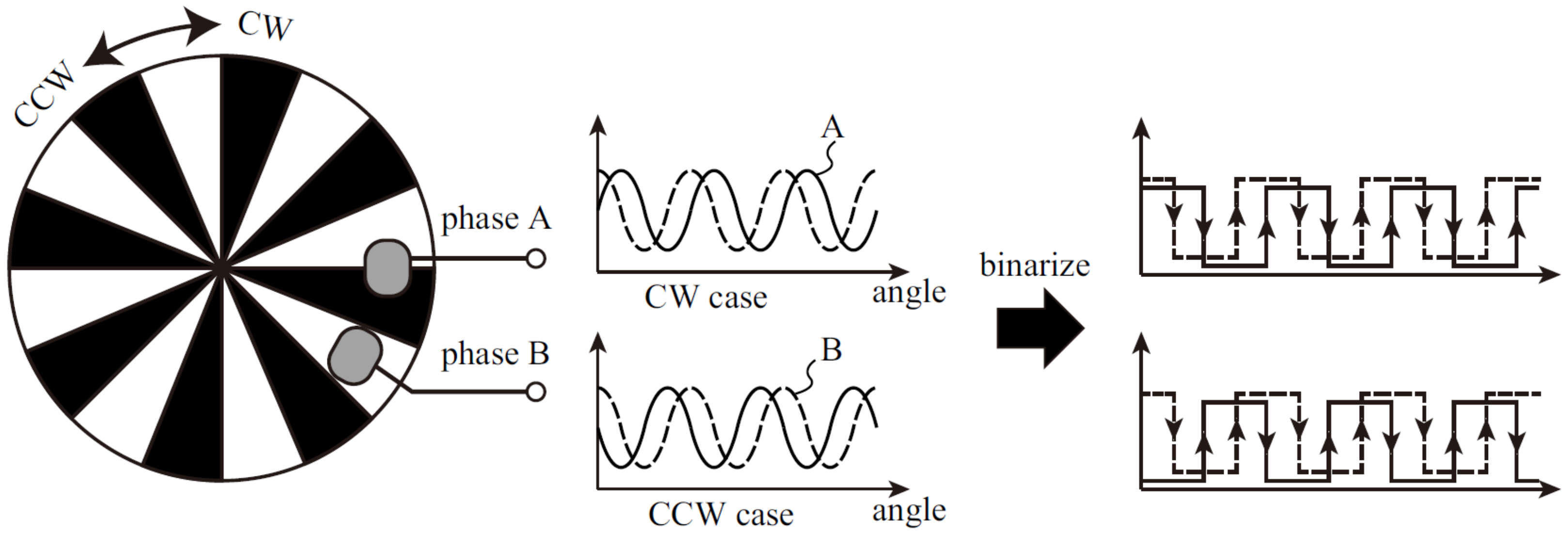




Add Encoder Feedback to Microcontroller

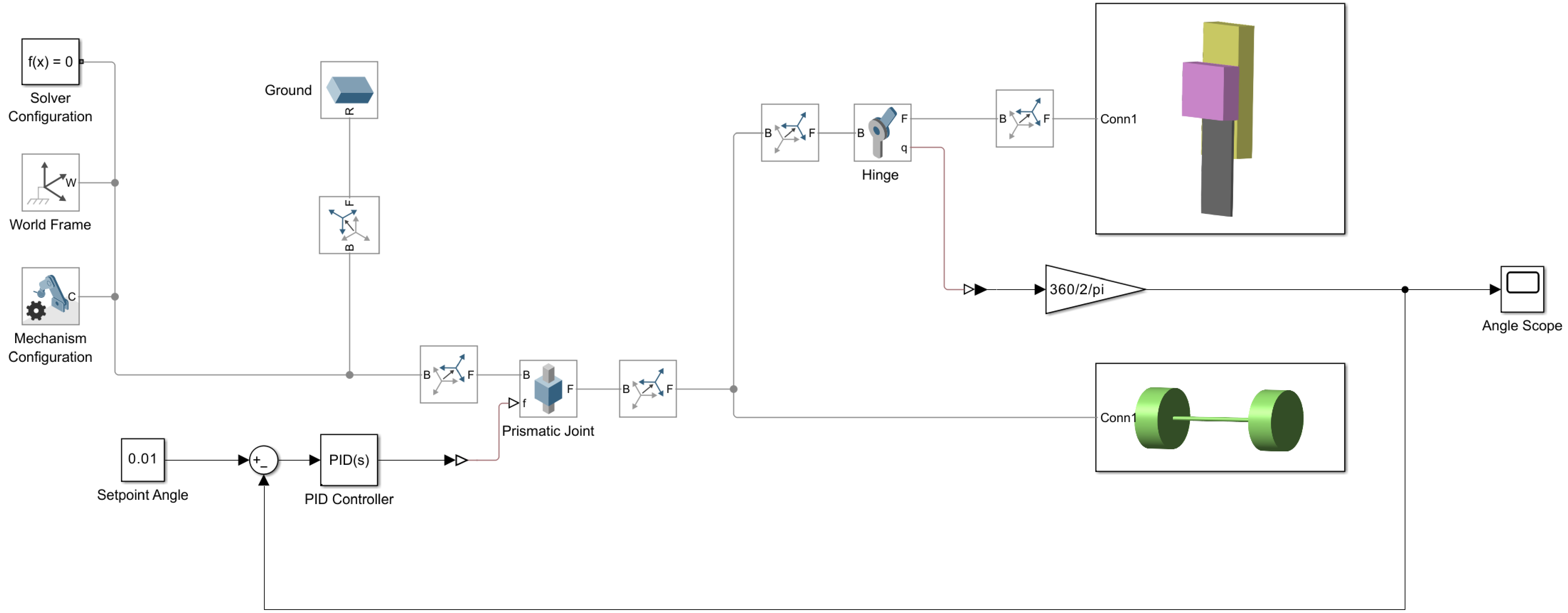

```c
// Sampling frequency
#define SAMP_FREQ 2000.0f
// Motor voltage limit (V_motor = 3.3 * v_out * 4)
#define VLIMIT 0.1f
// Motor offset voltage
#define VOFFSET 0.0f
// Feedback gains (Arbitrary Unit)
#define KP 1.0f
#define KD 0.0f
```

```c
// Motor output
if (pd1v < 0.02f && pd2v < 0.02f) {   // if not standing, stop the motor
    motor_v = 0;
    led_v_forward = 1;
    led_v_backward = 1;
} else {
    motor_v = v_out + VOFFSET;
    led_v_forward  = (v_dir > 0 ? v_out * 3.3f : 0.0f);
    led_v_backward = (v_dir < 0 ? v_out * 3.3f : 0.0f);
}
```

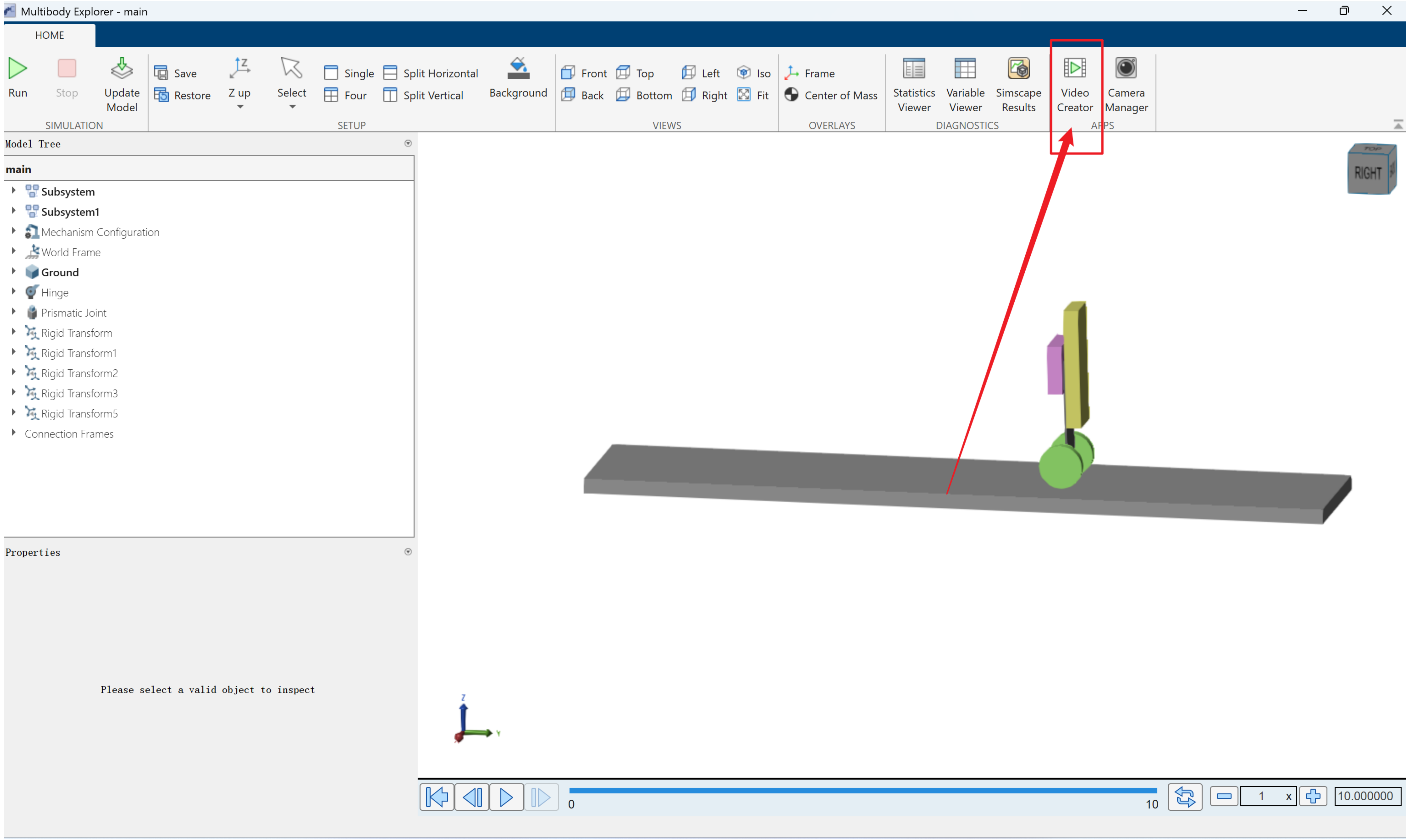


Use AI Tools to help you understanding the codes

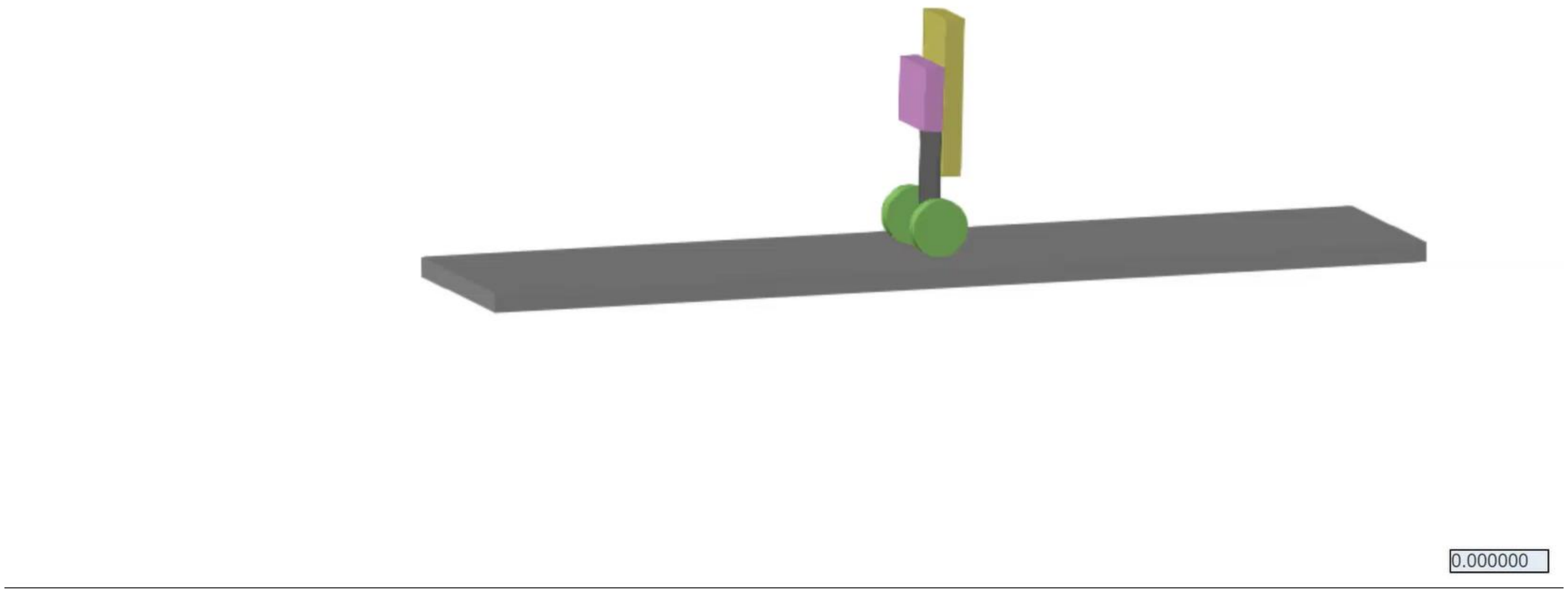
Simulink Simulation



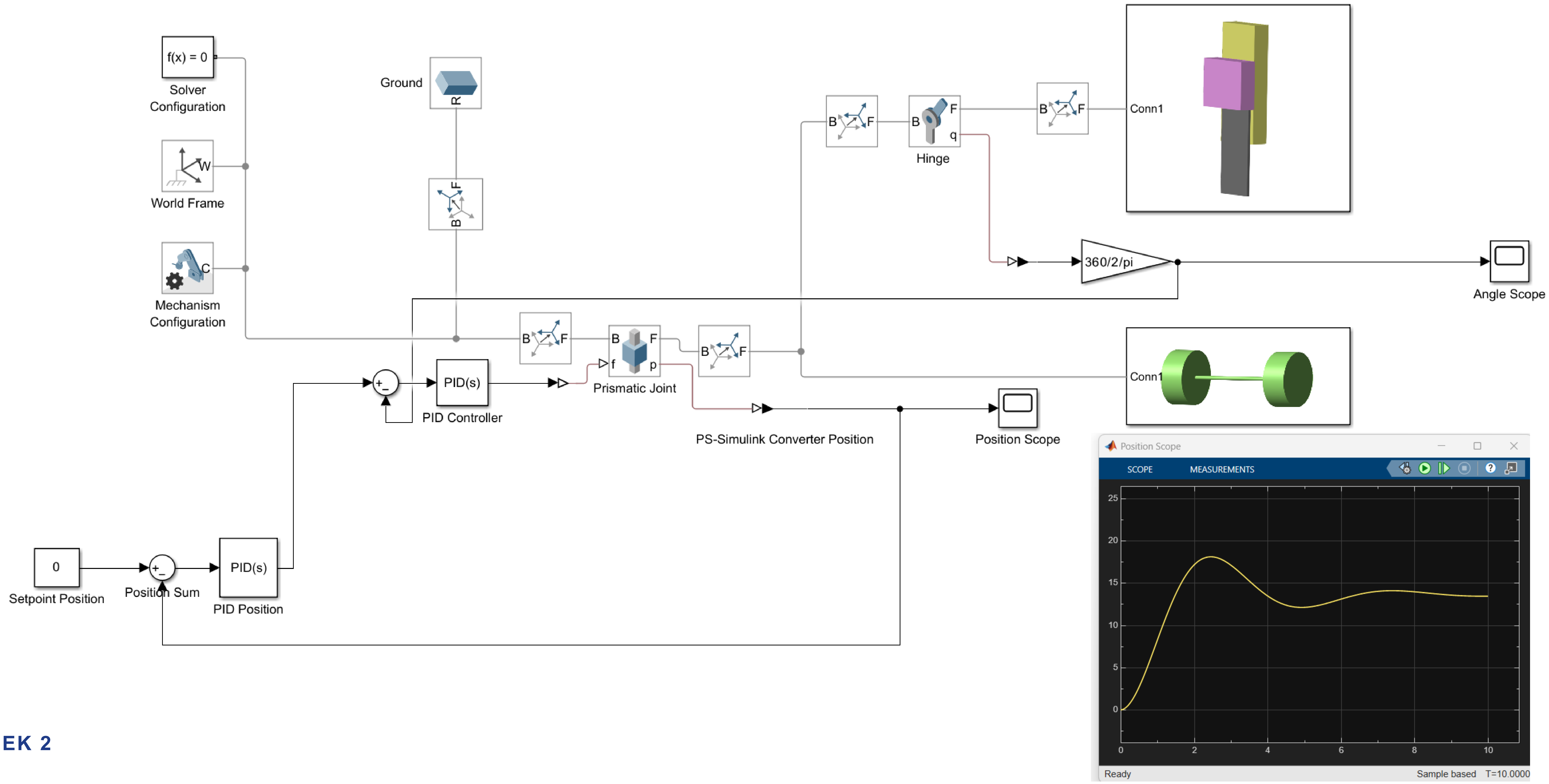
Simulink Simulation

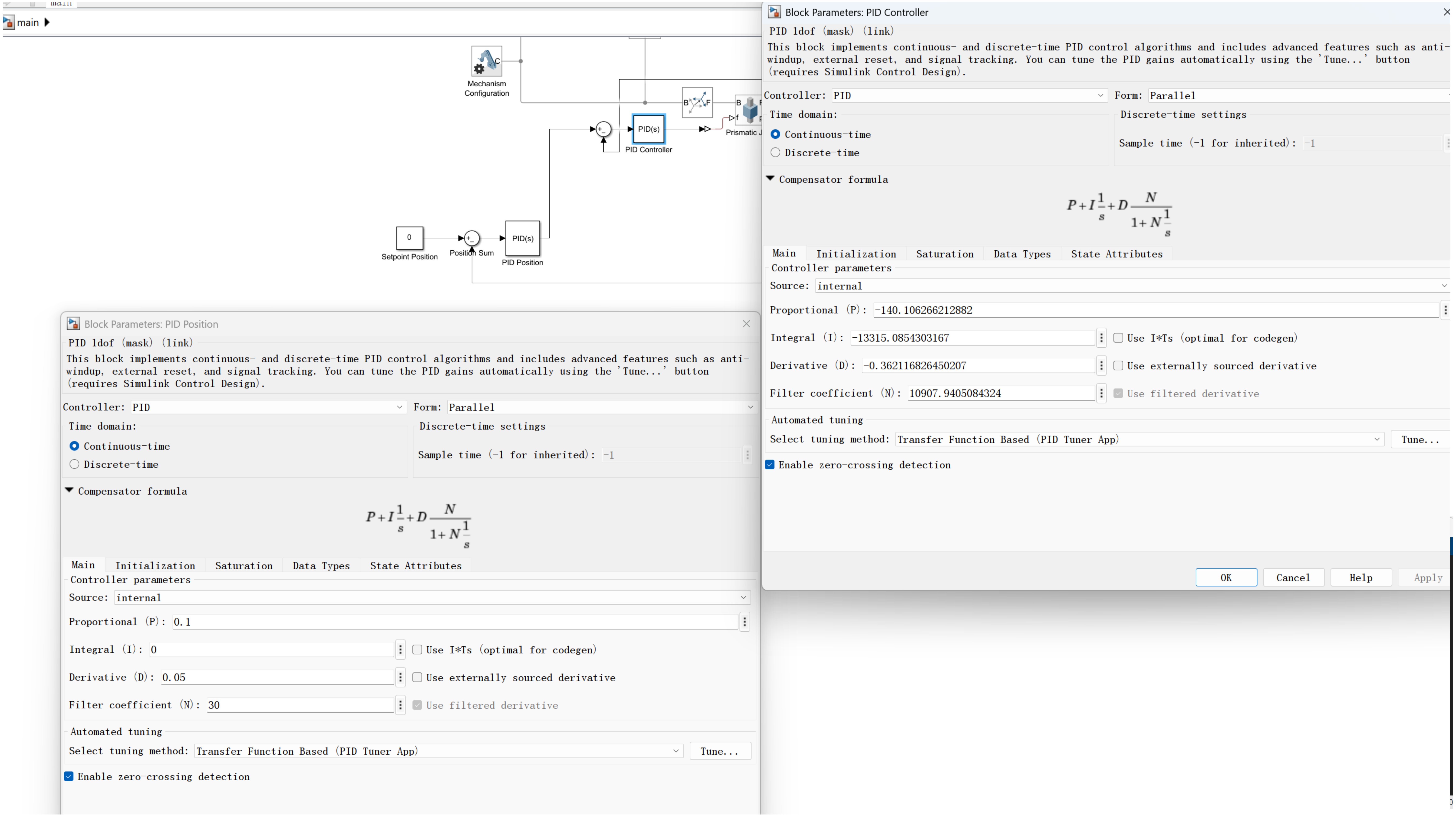


Simulink Simulation

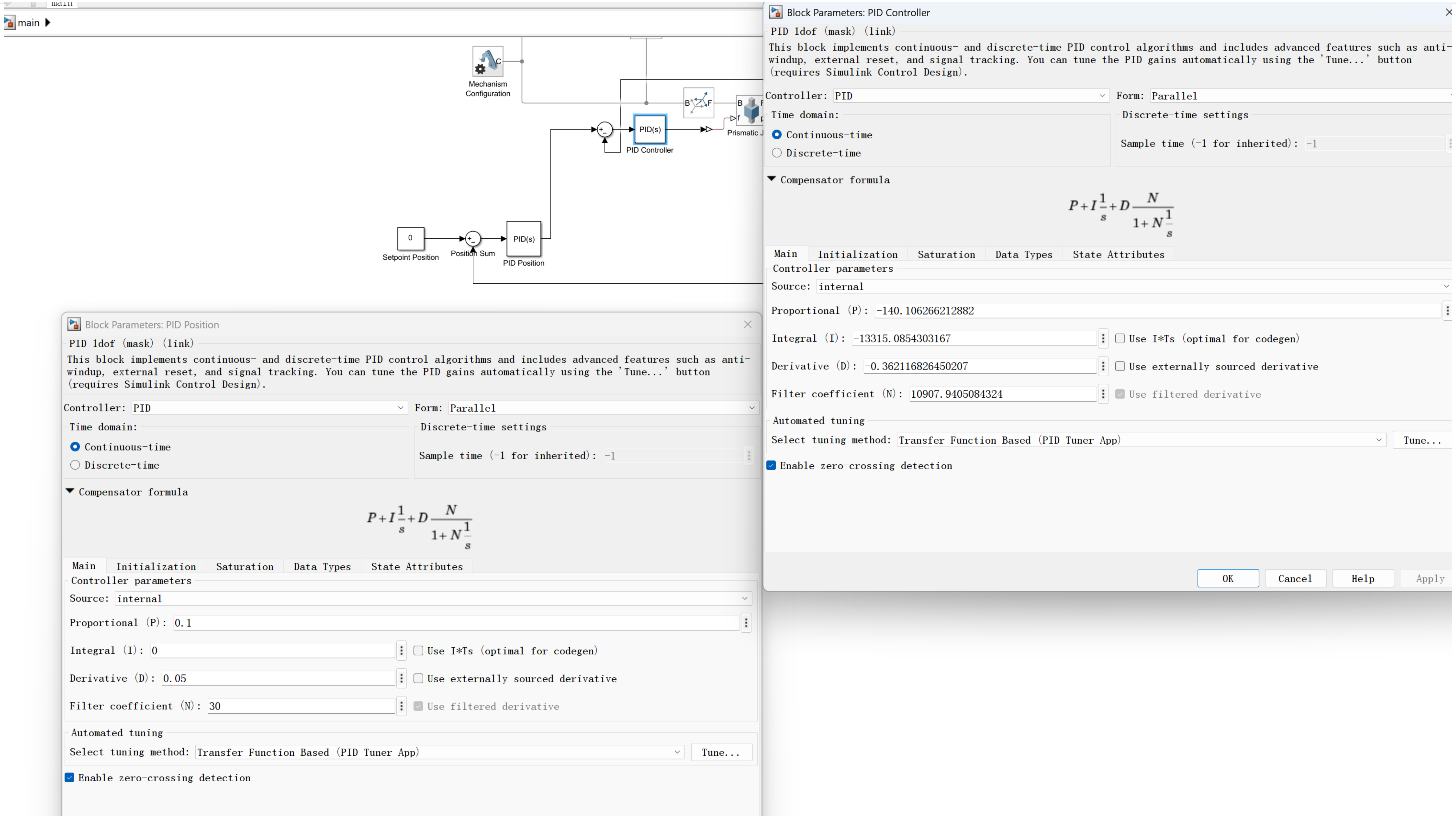


Simulink Simulation – Position Feedback

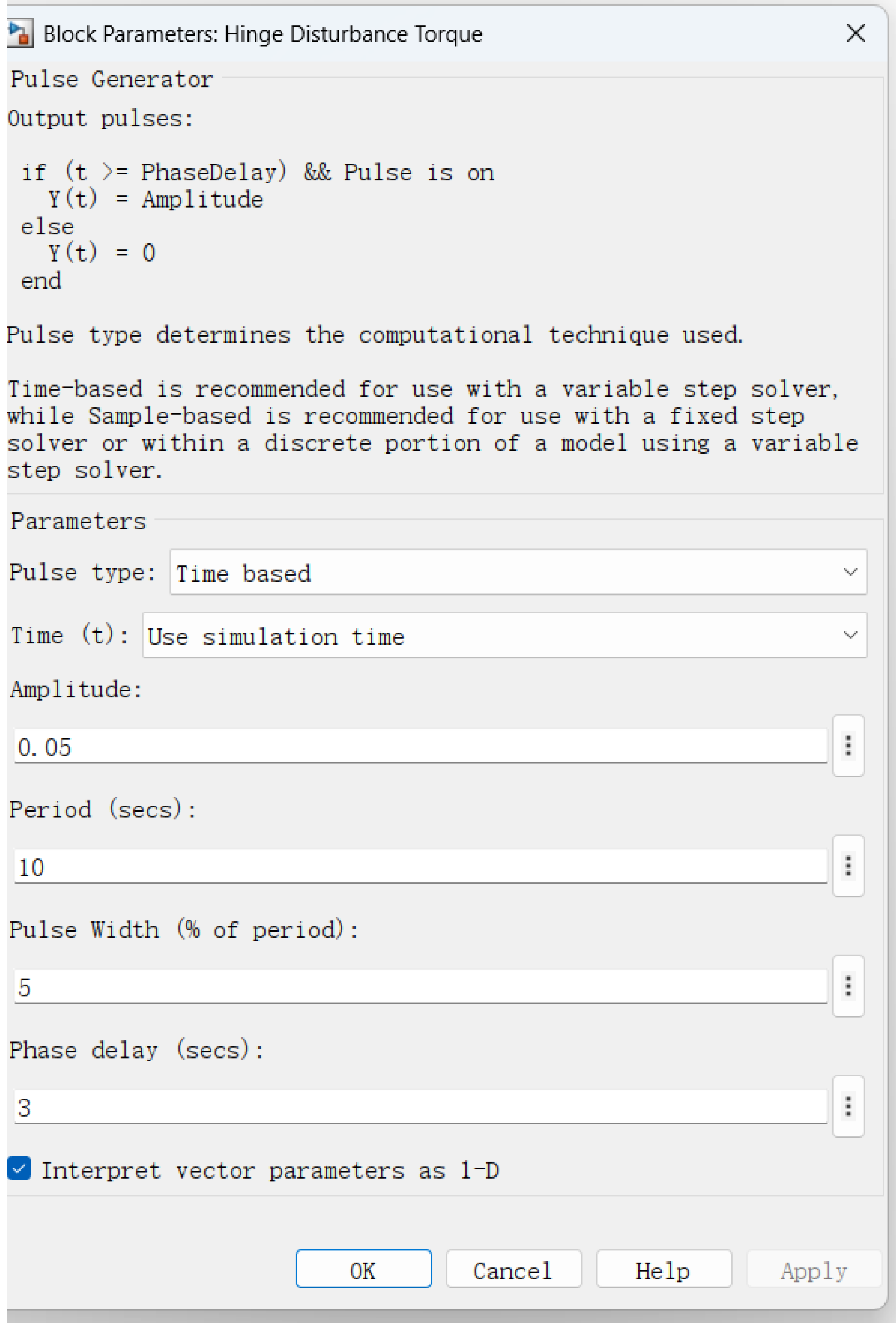


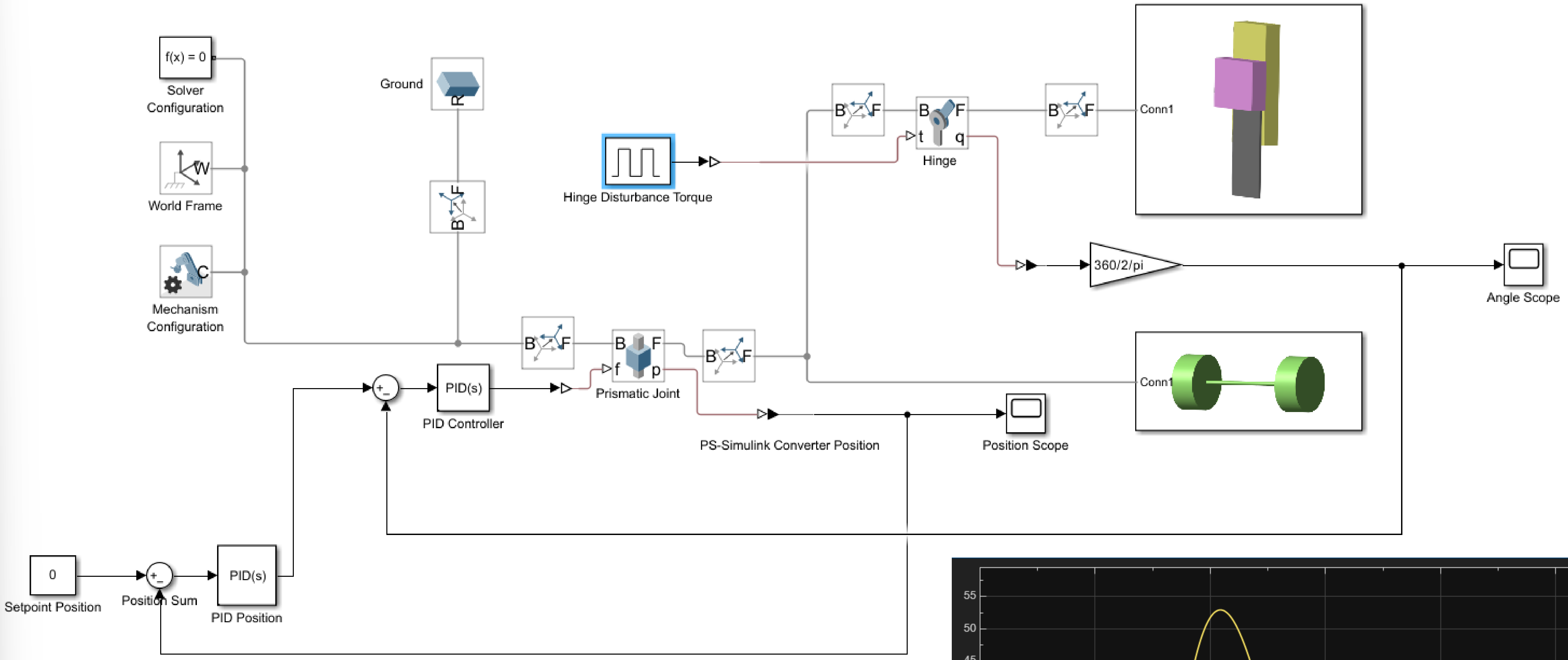

main
Mechanism Configuration
PID(s)
PID Controller
Prismatic J
0
Setpoint Position
Position Sum
PID(s)
PID Position
Block Parameters: PID Position
PID 1dof (mask) (link)
This block implements continuous- and discrete-time PID control algorithms and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).
Controller: PID
Form: Parallel
Time domain:
Continuous-time
Discrete-time
Discrete-time settings
Sample time (-1 for inherited): -1
Compensator formula
$P+I\frac{1}{s}+D\frac{N}{1+N\frac{1}{s}}$
Main
Initialization
Saturation
Data Types
State Attributes
Controller parameters
Source: internal
Proportional (P): 0.1
Integral (I): 0
Use I*Ts (optimal for codegen)
Derivative (D): 0.05
Use externally sourced derivative
Filter coefficient (N): 30
Use filtered derivative
Automated tuning
Select tuning method: Transfer Function Based (PID Tuner App)
Tune...
Enable zero-crossing detection
Block Parameters: PID Controller
PID 1dof (mask) (link)
This block implements continuous- and discrete-time PID control algorithms and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).
Controller: PID
Form: Parallel
Time domain:
Continuous-time
Discrete-time
Discrete-time settings
Sample time (-1 for inherited): -1
Compensator formula
$P+I\frac{1}{s}+D\frac{N}{1+N\frac{1}{s}}$
Main
Initialization
Saturation
Data Types
State Attributes
Controller parameters
Source: internal
Proportional (P): -140.106266212882
Integral (I): -13315.0854303167
Use I*Ts (optimal for codegen)
Derivative (D): -0.362116826450207
Use externally sourced derivative
Filter coefficient (N): 10907.9405084324
Use filtered derivative
Automated tuning
Select tuning method: Transfer Function Based (PID Tuner App)
Tune...
Enable zero-crossing detection
OK
Cancel
Help
Apply

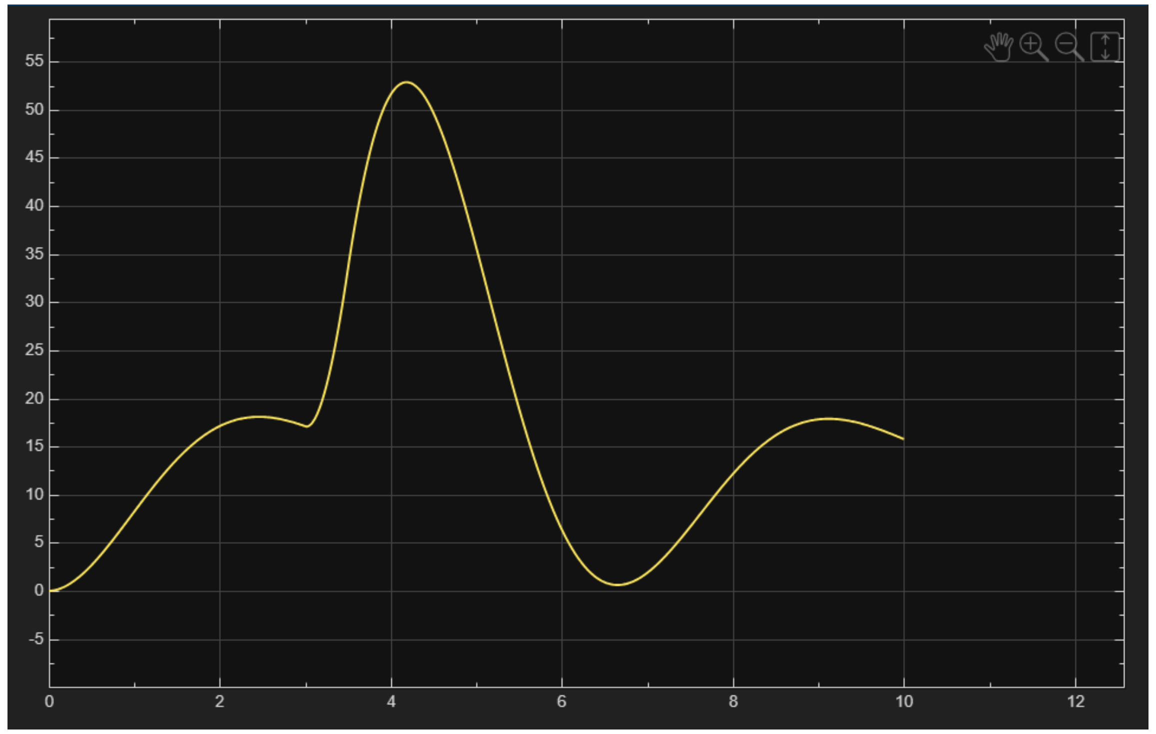
Simulink Simulation – Position Feedback



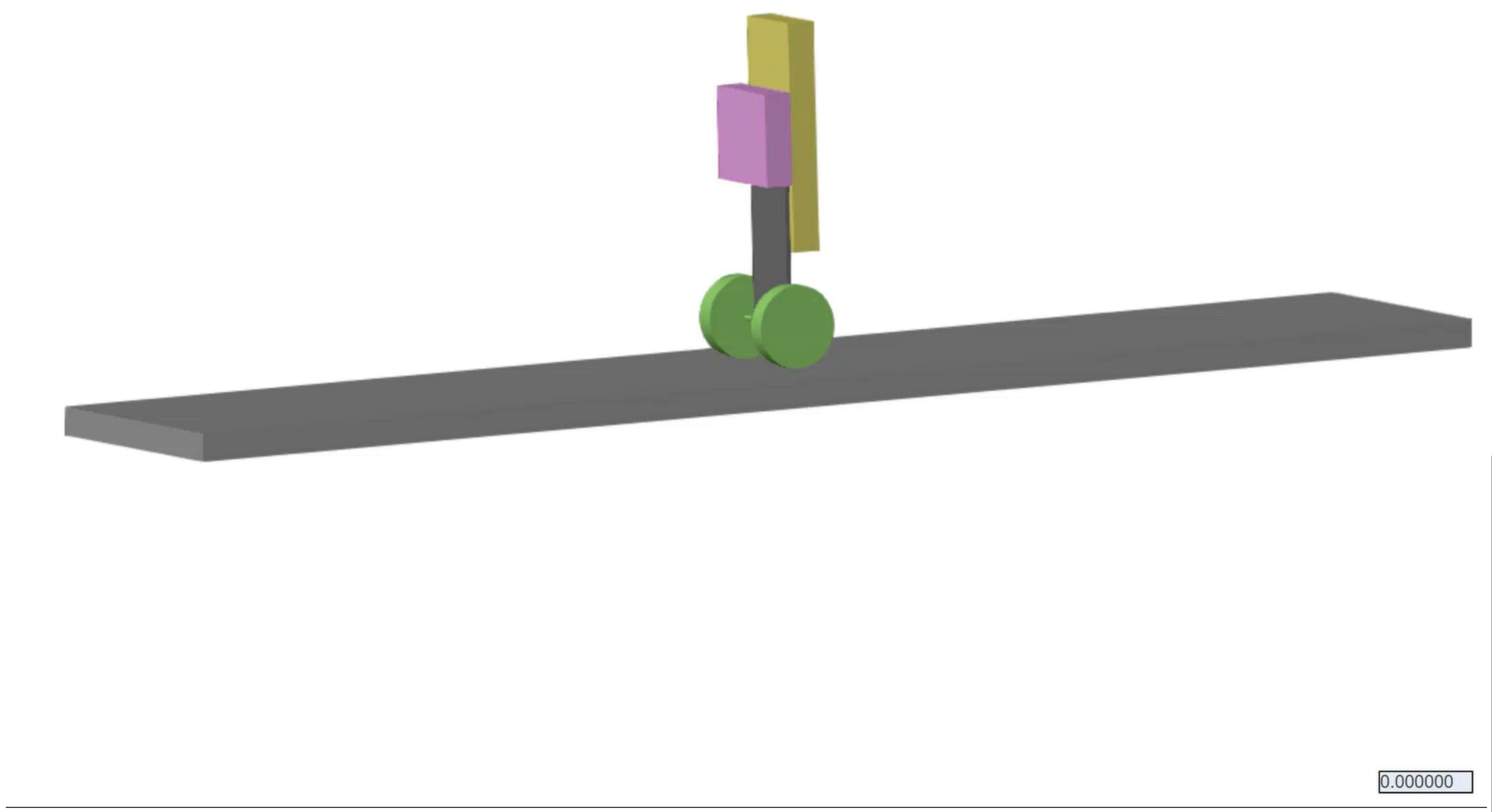
Simulink Simulation – Disturbance

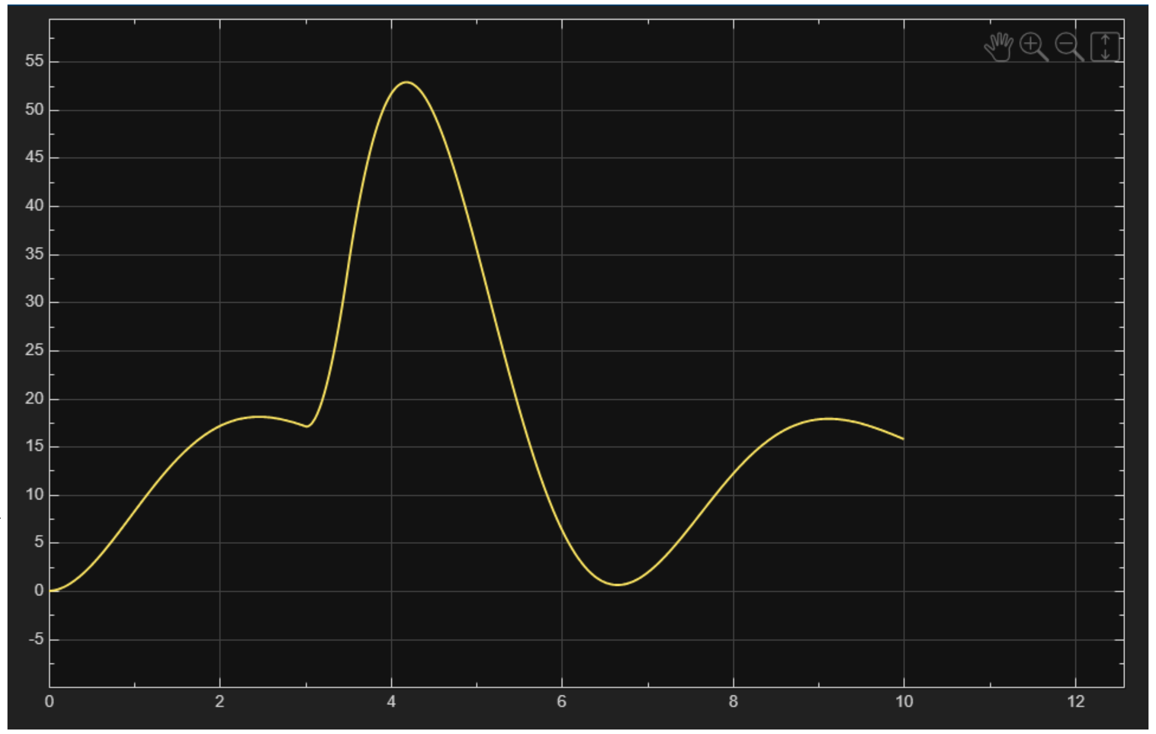






Simulink Simulation – Disturbance





simulink-agentic-toolkit

https://github.com/matlab/simulink-agentic-toolkit

How It Works

```
┌──────────┐       ┌──────────┐        ┌──────────┐
│ AI Agent │◄─MCP─►│MCP Server│◄──────►│ MATLAB / │
│ (Claude, │       │ (MATLAB  │        │ Simulink │
│  Cursor, │       │ MCP Core)│        └──────────┘
│  Copilot)│       └──────────┘
└──────────┘
      ▲
      │ reads
┌─────┴────┐
│  Skills  │
│ (MBD best│
│practices)│
└──────────┘
```

Your agent reads **skills** for domain knowledge, then calls **MCP tools** to interact with MATLAB and Simulink. The MATLAB MCP Core Server bridges the connection (downloaded during setup).

Supported Platforms

Platform	Setup	Notes
Claude Code	Automated	Also supports no-clone marketplace install (skills only)
GitHub Copilot	Automated	
OpenAI Codex	Automated	
Gemini CLI	Automated	
Sourcegraph Amp	Automated	
Cursor	Manual	Untested

Command Window

```
matlab.addons.install('C:\Users\rocke\.local\share\MATLABMCPCoreServerToolkit.mltbx')
addpath('D:\Local\20251230_Inverted_Pendulum\Week2\simulink-agentic-toolkit')
satk_initialize


ans =

  1×4 table

                Name                 Version    Enabled                 Identifier
    _____________________________    _______    _______    ______________________________________

    "MATLAB MCP Core Server Toolbox"  "0.1.0"    true      "fce99a56-1b63-4d49-9d70-c32c5605e4eb"


Simulink Agentic Toolkit - Installation Check
===============================================

Prerequisites
  ✓ MATLAB 25.2.0.3055257 (R2025b) Update 2
  ✓ Simulink installed

MATLAB Environment
  ✓ All 6 tool entry points on path
  ✓ Package resolution correct (+sage, +mbd inside SATK)

MCP Server
  ✓ matlab-mcp-core-server.exe found: C:\Users\rocke\.local\bin\matlab-mcp-core-server.exe

MCP Connectivity
  ✓ shareMATLABSession available (MATLABMCPCoreServerToolkit installed)
  ✓ MATLAB connector running (port 31516)

===============================================
Result: PASS
```

Week 2 Submission

If the robot cannot work yet, record its current state and clearly show which parts are already working.

Please also attach a PDF explaining the current status and the planned solution.

Video 1:

show the real wheeled inverted pendulum cart under a disturbance.

Video 2:

show the simulation running under normal conditions without an external disturbance.

Video 3:

show the simulation response when a disturbance is applied.

If the cart or simulation is not working perfectly yet, record its current state and clearly show which parts are already working.

DEADLINE

21:00

on the day of the class

Today — June 8

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Each group only needs one submission.